Progressive gate-unfreezing of the QFT operator

One gate thawed per stage, trained to a plateau — three unfreeze orderings × two initialisations on DIV2K-8q

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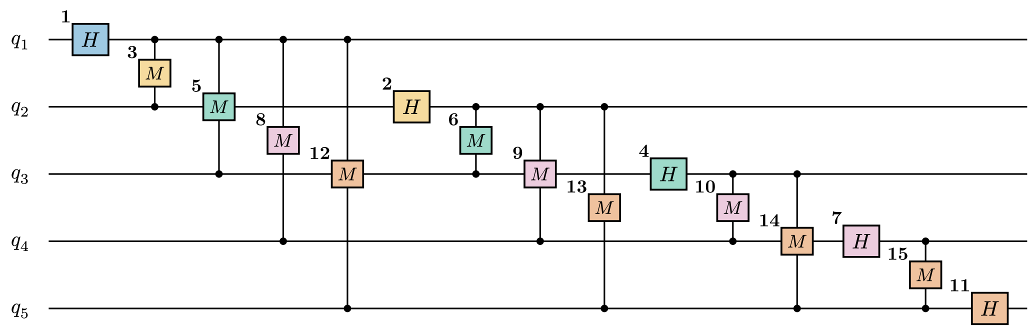
Question

The `QFTBasis(8, 8)` (72 U(2) gates) starts either at the QFT-family **identity** ($T_{t=0}(x) = x$) or **Haar-random**. We **unfreeze one gate per stage**, train it to a plateau, then thaw the next. Does the **order** — block-growth (`bg`), left→right (`lr`), right→left (`rl`) — or the **initialisation** change the trajectory or the endpoint?

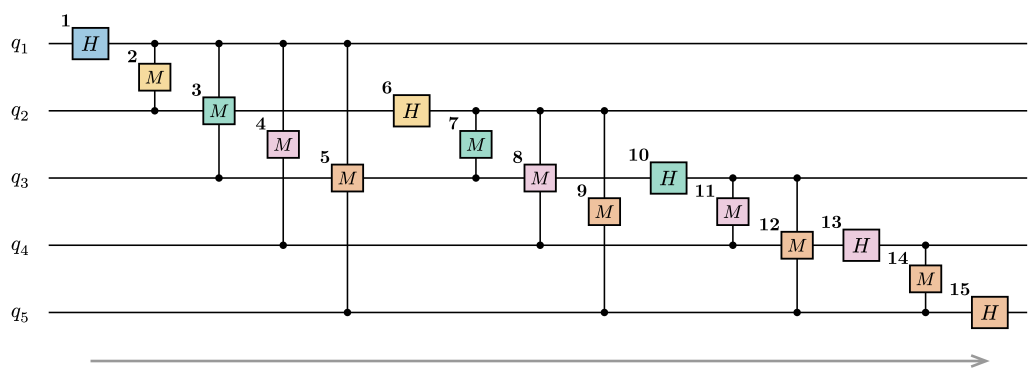
QFT gate-unfreezing — three orderings (single axis, m = 5)

Same QFT(5) circuit in the paper's tensor-network notation (controlled-phase M = box between two wires with legs to a dot on each). Bold badge = **unfreeze step** for that ordering; fill = block-stage **k** (= highest qubit a gate touches). Cumulative: each step thaws one more gate, the rest stay frozen at identity. (No SWAP gates — bit-reversal is folded into the index convention.)

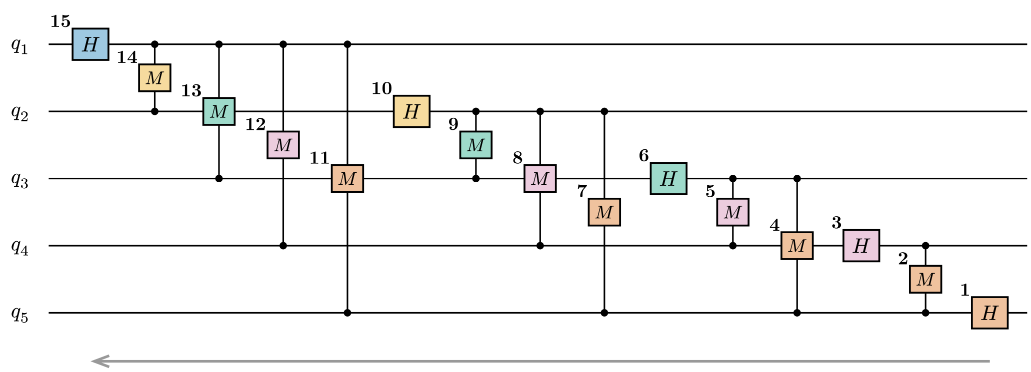
(a) block-growth stage k completes QFT(k, k); long-range couplings delayed — the gate-level mirror of the block sweep



(b) left → right sweep unfreeze in QFT construction order 1→15; q1 couples to all of q2–q5 before H_2 at step 6



(c) right → left sweep reverse construction order; last-built gate (H_5) first, H_1 last



Block-stage colours: k = 1 k = 2 k = 3 k = 4 k = 5

Figure 1: The three orderings on a QFT circuit (badge = unfreeze step, fill = block-stage k). **bg**: stage k completes QFT(k, k). **lr**: QFT construction order. **rl**: reverse. The same index sequences (at $m = 8$) drive every run.

Setup

Fixed batch of 50 training images; `MSELoss` on the top-10% coefficients. pdft's JIT Adam step runs with a per-stage frozen set (moments reset per stage); a Riemannian grad-norm probe gives the plateau signal. A stage ends at $\|g\| < 10^{-5}$ **or** $|\Delta L| < 10^{-5}$ (min 5 steps), else an 800-step cap. Random init: Haar $U(2)$ per Hadamard, uniform phase per controlled-phase, seed shared across orderings.

Training dynamics

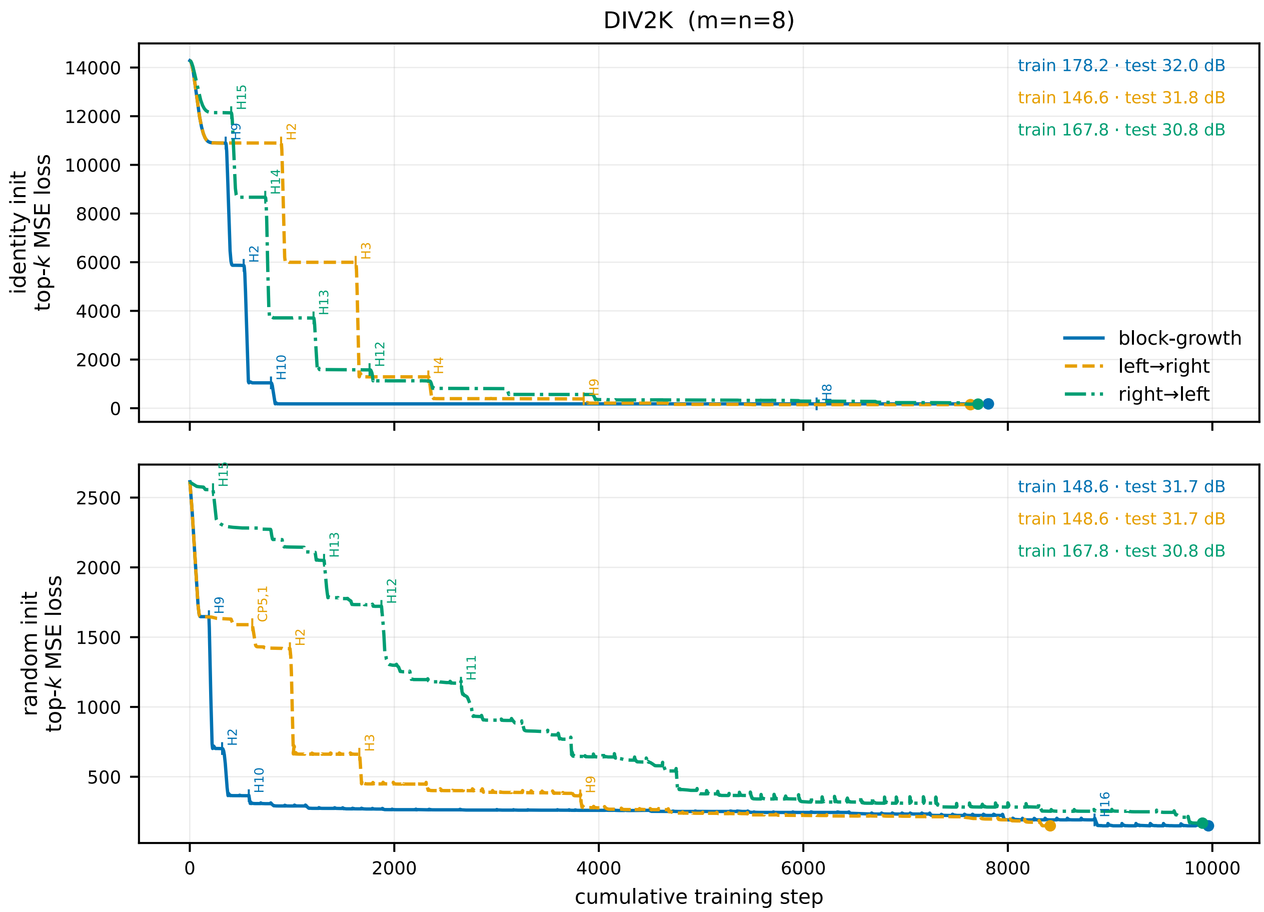


Figure 2: Absolute **training** top-k MSE loss vs cumulative step, one curve per ordering (rows: identity / random init). Tick labels mark **which gate** (e.g. `H7`, `CP3,1`) was thawed at each of the largest drops. Endpoints report **train MSE** and **test PSNR@ρ{=}.20** — note these are **different** metrics (training top-10%-coefficient loss vs test-set image reconstruction keeping 20%), so a curve can sit higher in train loss yet score a higher test PSNR. Per-cell loss+grad-norm staircases: `div2k_8q/<init>/figures/`.

Reconstruction PSNR (DIV2K)

Test PSNR (dB) at the final stage, with cumulative steps and plateau-trigger counts (grad-norm / loss-Δ / cap):

init	**order**	ρ{=}.05	ρ{=}.10	ρ{=}.15	ρ{=}.20	**steps (g/Δ/cap)**
identity	block-growth	19.4	26.9	29.7	32	7810 (0/72/0)
identity	left→right	25.2	27.9	30	31.8	7635 (0/72/0)
identity	right→left	24.8	27.2	29.1	30.8	7709 (0/72/0)
random	block-growth	25.2	27.8	29.8	31.7	9962 (0/72/0)
random	left→right	25.2	27.8	29.8	31.7	8415 (0/72/0)
random	right→left	24.8	27.2	29.1	30.8	9904 (0/72/0)

Block-size baseline (comparison)

`qft_progressive` reaches the **same** QFT$(8,8)$ on DIV2K via a **block-size** curriculum (train all $k(k+1)$ gates of a $2^k \times 2^k$ QFT at once, $k = 1..8$):

k	block	# gates	ρ\{=\}.05	ρ\{=\}.10	ρ\{=\}.15	ρ\{=\}.20
1	2×2	2	9.4	12.5	16.2	21.1
2	4×4	6	19.1	26.9	29.8	32
3	8×8	12	25.2	28.1	30.3	32.3
4	16×16	20	25.3	28	30.1	32
5	32×32	30	25.2	27.8	29.8	31.7
6	64×64	42	25.2	27.8	29.8	31.7
7	128×128	56	25.2	27.8	29.8	31.7
8	256×256	72	25.2	27.8	29.8	31.7

It saturates by $k = 2$; the full QFT$(8,8)$ endpoint is **31.7 dB** @ρ\{=\}.20. The gate-unfreeze sweep lands at the same value (identity `bg/lr` 32/31.8 dB, random ≈31.7), so the QFT$(8,8)$ optimum is **schedule-independent**. For reference, the best **any** prior trained basis reaches on DIV2K is 33.7 dB (`#pastmax.div2k_8q.by_rho.at("0.20").method`); the QFT family sits ≈2 dB under the richer U(4) families.

Reading

The staircase shows each gate's marginal value: a tall drop is a gate that mattered (mostly the early, small-block gates — see the marked steps), a flat plateau a gate left near where it was thawed. All three orderings and both inits converge to ≈31.7 dB, so **order and curriculum set only the path, not the destination** — `rl` merely trails by ≈1 dB en route. The train-loss order need not match the test-PSNR order (`bg` has the **highest** train MSE yet the **best** test PSNR): the loss is a top-10%-coefficient proxy on the train batch, PSNR a 20%-keep reconstruction on the test set, so the two rank slightly differently in this near-flat optimum.